

Human giant GTPase GVIN1 forms an antimicrobial coatomer around the intracellular bacterial pathogen *Burkholderia thailandensis*

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eLife Assessment

This is a **fundamental** discovery revealing two independent IFN γ -induced pathways that restrict bacterial motility: one GBP1-dependent and the other GVIN1-dependent. The findings are supported by **compelling** evidence. While the paper is already very strong, there are a few points that could be addressed editorially or through the addition of a few key experiments.

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Abstract

Several human pathogens exploit the kinetic forces generated by polymerizing actin to power their intracellular motility. Human cell-autonomous immune responses activated by the cytokine interferon-gamma (IFN γ) interfere with such microbial actin-based motility, yet the underlying molecular mechanisms are poorly defined. Here, we identify the IFN γ -inducible human giant GTPases GVIN1 as a novel host defense protein that blocks the bacterial pathogen *Burkholderia thailandensis* from high-jacking the host's actin polymerization machinery. We found that GVIN1 proteins form a coatomer around cytosolic bacteria and prevent *Burkholderia* from establishing force-generating actin comet tails. Coatomers formed by a second IFN γ -inducible GTPase, human guanylate binding protein 1 (GBP1), constitute a GVIN1-independent but mechanistically related anti-motility pathway. We show that coating with either GVIN1 or GBP1 displaces the *Burkholderia* outer membrane protein BimA, an actin nucleator that is essential for actin tail formation. Both GVIN1 and GBP1 coatomers require additional IFN γ -inducible co-factors to disrupt the membrane localization of BimA, demonstrating the existence of two parallel-acting IFN γ -inducible defense modules that evolved to target a virulence trait critical for the pathogenesis of numerous bacterial infectious agents.

Introduction

Actin-based bacterial motility is a key mechanism by which host cytosol-invading bacterial pathogens like *Listeria*, *Shigella*, and *Burkholderia* move within and between infected cells. By recruiting and polymerizing host actin filaments at the surface of one bacterial pole, these bacteria form actin “comet tails” that propel the bacteria forward, enabling them to move rapidly through the infected cell and directly into neighboring cells (1, 2). This dissemination mechanism enables pathogens to spread within cell layers while avoiding extracellular immunity. Intracellular cell-autonomous host defenses that target actin motility are thus critical for immune protection, yet our understanding of these defenses is limited.

The general mechanisms by which intra-cytosolic bacterial pathogens hijack the host cell’s actin polymerization machinery are well characterized. For example, in *Shigella* the outer membrane autotransporter IcsA is localized to one bacterial pole and recruits the host protein N-WASP, an actin nucleation promotion factor. N-WASP in turn recruits the actin polymerization initiation Arp2/3 complex to launch actin filament formation (3–7). Like IcsA in *Shigella*, the *Burkholderia* outer membrane protein BimA is localized to one bacterial pole and is essential for actin-based motility (8, 9). BimA of *Burkholderia thailandensis* is an N-WASP mimic that directly recruits Arp2/3 to the bacterial pole to initiate actin polymerization (10).

The polar localization of bacterial proteins like IcsA and BimA is essential for their function and tightly regulated through multiple mechanisms. Specifically, the function of IcsA is dependent on lipopolysaccharide (LPS), the main building block of the outer leaflet of Gram-negative bacterial outer membranes. LPS consists of its membrane anchor lipid A, an inner core oligosaccharide, and the outer core polysaccharide O-antigen. The outwards-facing thick O-antigen layer provides a chemo-physical barrier that protects bacteria against antimicrobial molecules (11). Additionally, O-antigen enhances bacterial outer membrane stiffness (12). Accordingly, loss of O-antigen in ‘rough mutant’ bacteria results in increased outer membrane fluidity and enhanced diffusion of outer membrane proteins (12). The functional importance of this O-antigen-mediated membrane stiffness is illustrated by the enhanced diffusion of IcsA proteins away from the bacterial pole and the consequential loss of actin-based motility in *Shigella flexneri* rough mutants (13–15). Whether O-antigen is also required for the function of the actin nucleator BimA of *Burkholderia* has not been previously reported.

Actin-based cell-to-cell movement not only facilitates bacterial spread within tissues, but also helps pathogens avoid extracellular immunity. To control infections with pathogens that largely remain hidden intracellularly, host cells are equipped with cell-autonomous immune responses. These cell-autonomous defense programs are executed by a range of immune effector proteins, many of which are expressed in response to proinflammatory cytokine signaling. Arguably, the most potent inducer of antibacterial cell-autonomous immunity is the lymphocyte-derived cytokine IFN γ (16). Among the hundreds of IFN γ -inducible genes are those encoding members of GTPase families that include ‘Immunity-Related GTPases’ (IRGs), giant GTPases annotated as ‘GTPases, Very large Interferon-inducible’ (GVINs), and ‘Guanylate Binding Proteins’ (GBPs). Whereas GBPs and other interferon-inducible GTPases families have proven antimicrobial activities, no biological function has previously been reported for GVIN proteins (17).

The most highly expressed human GBP family member is GBP1, an LPS-binding protein that can form coatomers on the surface of Gram-negative bacteria including *Shigella* and *Burkholderia* (18–20). Once embedded onto the bacterial outer membrane, the GBP1 coatomer operates akin to a surfactant that disrupts the function of O-antigen (13). As a result, IcsA is displaced from the bacterial pole of GBP1-coated bacteria (13), which consequently lose their ability to form actin tails and to disseminate within epithelial monolayers (13, 21, 22). Human GBP1

coatamers also assemble on the surface of *B. thailandensis* (21 [↗](#)). However, studies conducted in human HeLa epithelial cells indicated that these GBP1 coatamers formed on the surface of *B. thailandensis* fail to disrupt the formation of actin tails. Instead, it was suggested that GBP1-mediated host cell death may inhibit the intercellular spread of *B. thailandensis* (23 [↗](#)). In contrast to these findings in human HeLa cells, a separate study demonstrated that the mouse ortholog of human GBP1, mouse Gbp2, was capable of directly interfering with *B. thailandensis* actin tail formation in murine primary macrophages (24 [↗](#)). However, the underlying mechanism by which mouse Gbp2 modulates *Burkholderia*'s actin-based motility was not further explored.

In the current work we demonstrate that some human cell lines block *B. thailandensis* actin tail formation when primed with IFN γ , whereas other human cell lines including commonly used HeLa cervical epithelial cells lack the ability to interfere with *B. thailandensis* actin tail formation. We find that IFN γ priming can activate two independent but functionally related anti-motility pathways that are dysfunctional in HeLa cells but functional in T24 bladder epithelial cells. The first pathway requires the formation of GBP1 coatamers and the expression of additional IFN γ -inducible factors that are ostensibly non-functional or not expressed in HeLa cells. A second anti-motility pathway is dependent on the expression of GVIN1, the single human member of the GVIN family of IFN-inducible giant GTPases (17 [↗](#)). We show that GVIN1, annotated as the pseudogene GVINP1 (GTPase, Very large IFN-Inducible Pseudogene-1) in the NCBI database, is a functional protein that forms a coamer surrounding cytosolic *B. thailandensis*. Like GBP1 coatamers, GVIN1 coatamers require additional IFN γ -inducible factors to disrupt the localization of BimA and to interfere with actin tail formation. Our study thus identifies GVIN proteins as a novel class of antimicrobial coamer-forming proteins.

Results

IFN γ priming restricts *B. thailandensis* actin tail formation in a subset of human cell lines

Confirming previous observations (13 [↗](#), 21 [↗](#), 22 [↗](#)), we found that GBP1 coatamers form on the surface of *S. flexneri* in IFN γ -primed human HeLa and T24 epithelial cell lines, prevent recruitment of Arp2/3, and block bacterial actin-tail formation (Fig. S1A-G [↗](#)). Although GBP1 coatamers also form on the surface of *B. thailandensis* in IFN γ -primed HeLa cells (Fig. 1A-B [↗](#)), such GBP1 coatamers failed to diminish Arp2/3 recruitment to the bacterial pole (Fig. S1H-I [↗](#)) and had no discernable impact on the percentage of bacteria that formed actin tails (Fig. 1C [↗](#)). These data confirmed that IFN γ -primed HeLa cells block actin tail formation by *S. flexneri* but not by *B. thailandensis*.

To more broadly test whether any human cell lines exist that could block actin tail formation by *B. thailandensis*, we monitored the effect of IFN γ priming on actin tail formation in a small collection of commonly used human cell lines that included A549 lung epithelial cells, T24 bladder epithelial cells, and differentiated monocytic U937 and THP1 cells. In the case of THP1 monocytes we used CASP4-deficient (CASP4^{KO}) cells to avoid the confounding effects of *B. thailandensis*-triggered CASP4-driven pyroptotic cell death. We observed that IFN γ priming significantly diminished actin tail formation by *B. thailandensis* in T24, U937, and THP1-CASP4^{KO} but not in A549 cells (Fig. 1D-F [↗](#) and S2A-C [↗](#)). Deletion of GBP1 in IFN γ -primed epithelial T24 cells partially restored actin tail formation by *B. thailandensis* (Fig. 1D [↗](#) and 1F [↗](#)), indicating that both GBP1-dependent and GBP1-independent pathways impede *B. thailandensis* actin tail formation in human T24 cells. The GBP1-independent pathway is ostensibly inactive against *S. flexneri* based on the observation that the reduction in actin tail formation by *S. flexneri* in IFN γ -primed T24 cells is completely reversed by the deletion of GBP1 (Fig. S1F-G [↗](#)).

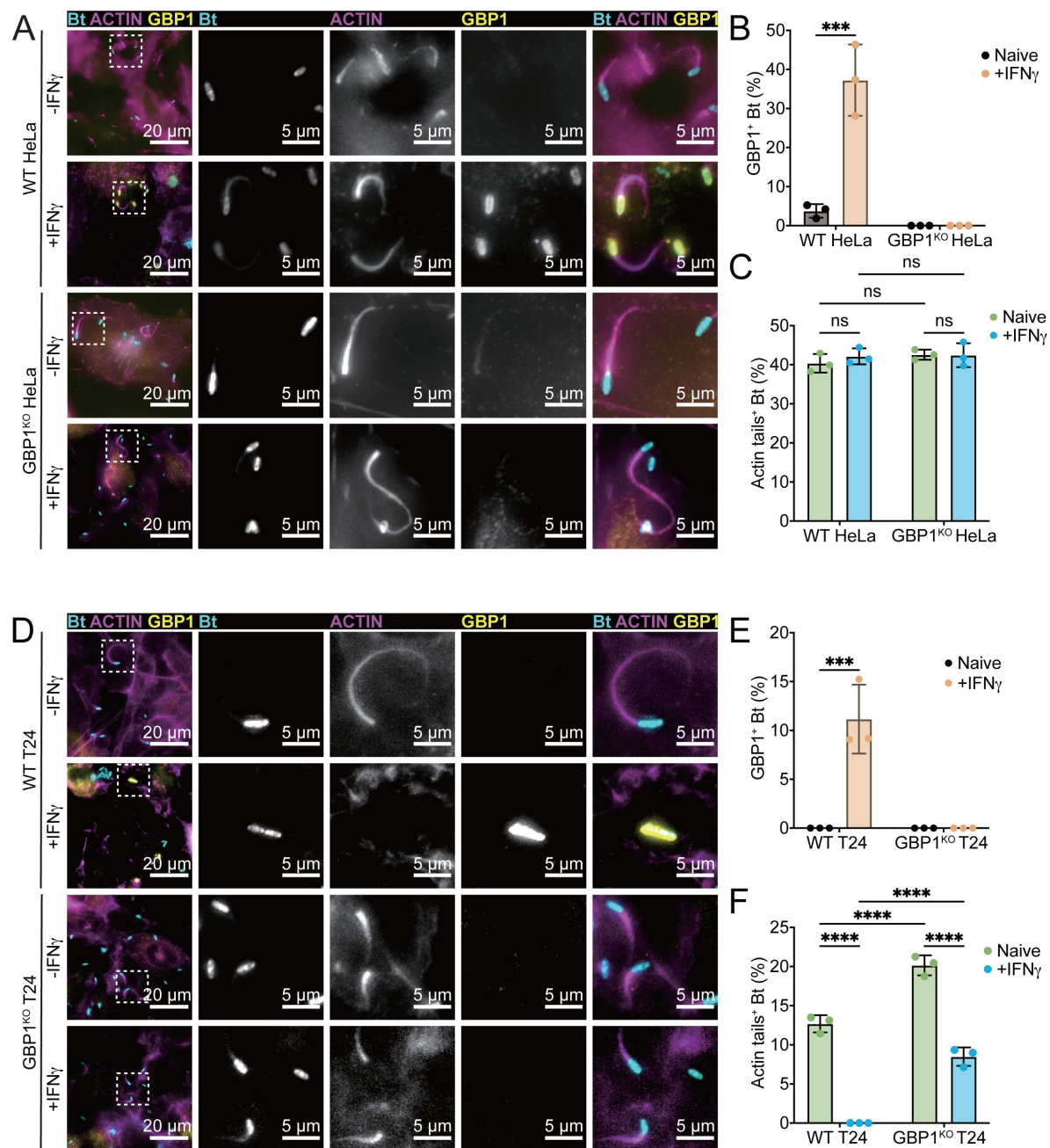


Fig. 1.

IFN γ -primed T24 cells restrict actin tails formation by *B. thailandensis*.

(A-C) Naïve or 200 U/mL IFN γ primed WT or GBP1^{KO} HeLa cells were infected with WT *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for GBP1 and actin. Percentages of GBP1-positive *B. thailandensis* (B) and bacteria with actin tails (C) were quantified. (D-F) Naïve or 200 U/mL IFN γ primed WT or GBP1^{KO} T24 cells were infected with WT *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for GBP1 and Actin. Percentages of GBP1-positive *B. thailandensis* (E) and bacteria with actin tails (F) were quantified. All bar graphs show 3 independent biological replicates, presented as mean $\pm$ standard deviation (SD). Two-way ANOVA with Tukey's multiple comparison tests were performed, with specific p-values indicated as follows: ***, p < 0.001; ****, p < 0.0001; ns, not significant.

Previous work suggested that GBP1 may limit cell-to-cell spread by *B. thailandensis* through the induction of pyroptotic cell death in HeLa cells (23). However, such a mechanism is unlikely to be active in T24 cells, because we found that naïve and IFN γ -primed WT and GBP1^{KO} T24 cells displayed comparable levels of cell death as assessed by LDH release assays (Fig. S2D).

GBP1-mediated inhibition of *B. thailandensis* actin tail formation requires a co-factor that is missing from HeLa cells

HeLa cells lack not only the GBP1-independent but also the GBP1-dependent pathway to restrict *B. thailandensis* actin tail formation, despite having GBP1 protein expression levels that are comparable to T24 cells (Fig. 2A). We therefore considered that the loss of the GBP1-dependent pathway in HeLa cells was either due to i) differences in the GBP1 protein sequence between the two cell lines or ii) the absence of an essential co-factor from HeLa cells. To test both hypotheses in one experiment, we engineered GBP1^{KO} T24 and GBP1^{KO} HeLa cells that inducibly expressed the same N-terminally mCherry-tagged GBP1 (mCh-GBP1) fusion protein upon treatment of cells with anhydrotetracycline (aTc) (Fig. 2B). Targeting of mCh-GBP1 to *B. thailandensis* occurred at comparable frequency in GBP1^{KO} T24 and GBP1^{KO} HeLa cells (Fig. S2E-F). We found that mCh-GBP1 overexpression was sufficient to reduce the percentage of actin tail-positive *B. thailandensis* by 2-fold in unprimed GBP1^{KO} T24 but not in unprimed GBP1^{KO} HeLa cells (Fig. 2C-D). These data suggest that T24 but not HeLa cells express an essential GBP1 co-factor, even in the absence of IFN γ priming. Combining ectopic mCh-GBP1 expression with IFN γ priming led to a near complete elimination of actin tail formation in T24 GBP1^{KO} but had no discernable effect on bacterial actin tail formation in HeLa GBP1^{KO} cells (Fig. 2C-D). The additive effect of ectopic GBP1 expression and IFN γ priming in T24 GBP1^{KO} cells can be largely explained by the induction of a GBP1-independent host defense pathway in IFN γ -primed T24 cells. However, IFN γ stimulation may additionally enhance GBP1-dependent host defense against *B. thailandensis*, if the putative GBP1 co-factor is itself an IFN γ -inducible gene (ISG).

Loss of O-antigen renders *B. thailandensis* resistant to GBP1-independent inhibition of actin tail formation

GBP1 coatomer formation is sufficient to block actin formation by *S. flexneri* but not by *B. thailandensis* (Fig. S1 and 2 and (13, 21, 22)). The mechanism by which GBP1 coatomers block actin tail formation was previously elucidated for *S. flexneri*: GBP1 coatomers on the surface of *S. flexneri* emulsify the O-antigen layer and thereby turn bacteria into quasi-O-antigen deficient strains. Because O-antigen is essential for the polar localization and function of IcsA (13–15), formation of GBP1 coatomers on *S. flexneri* is sufficient to block IcsA-dependent actin tail formation (13). To explain why GBP1 coatomer formation on the other hand is insufficient to block actin formation by *B. thailandensis*, we considered two hypotheses: i) the O-antigen layer of *B. thailandensis* is impervious to disruption by GBP1 coating or ii) O-antigen is dispensable for *B. thailandensis* BimA function. To test the latter hypothesis, we generated loss-of-function mutants in the gene BTH_I1483, which encodes the predicted polysaccharide biosynthetic enzyme WbpM (Fig. 3A). Immunofluorescence staining with *B. thailandensis*-specific anti-LPS antibody and ProQ Emerald 300 LPS gel staining characterized the *wbpM* CmR insertion mutant (henceforth simplified as $\Delta wbpM$) as O-antigen-deficient (Fig. 3B-C). We found that the O-antigen-deficient *wbpM* mutant formed actin tails at frequencies comparable to wildtype (WT) bacteria in naïve T24 cells (Fig. 3D-E), thus demonstrating that O-antigen is dispensable for *B. thailandensis* actin-based motility. These findings suggest that BimA remains functional even when the membrane stiffening property of O-antigen is disrupted by GBP1 coatomers. Unexpectedly, we also observed that the O-antigen deficient *B. thailandensis wbpM* mutant became resistant to IFN γ -mediated inhibition of bacterial actin tail formation in T24 GBP1^{KO} cells (Fig. 3E). The latter finding suggested that the GBP1-independent host defense pathway specifically recognizes the O-antigen polysaccharide layer of the bacterial outer membrane.

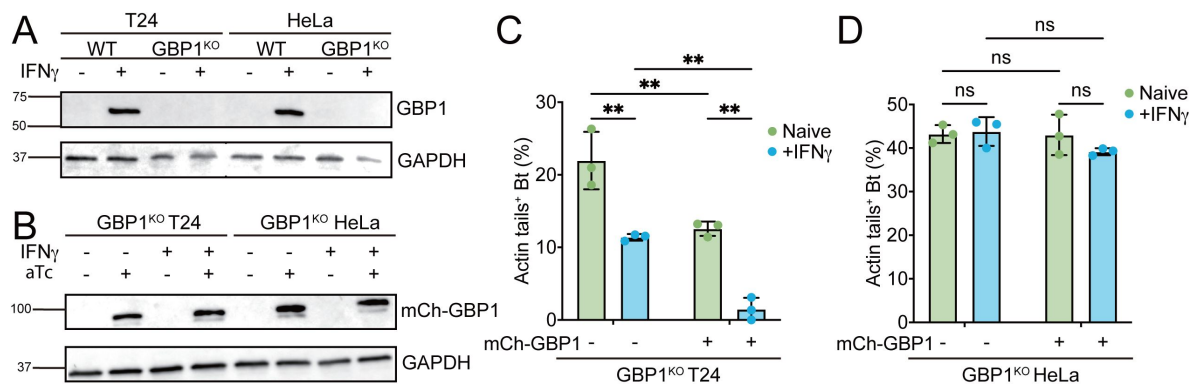


Fig. 2.

GBP1 requires a co-factor to restrict *B. thailandensis* actin tail formation.

(A) Immunoblotting for GBP1 protein expression in WT HeLa, GBP1^{KO} HeLa, WT T24, or GBP1^{KO} T24 cells treated with or without 200 U/ml IFN γ . (B-D) Parental GBP1^{KO} HeLa or GBP1^{KO} T24 cells were transduced with an anhydrotetracycline (aTc) - inducible mCherry-GBP1 lentiviral construct (mCh-GBP1). (B) Immunoblotting for GBP1 in naïve or 200 U/mL IFN γ primed GBP1^{KO} HeLa + mCh-GBP1 or GBP1^{KO} T24 + mCh-GBP1 cells treated with or without 1 μ g/mL aTc for overnight. (C-D) Naïve or 200 U/mL IFN γ primed GBP1^{KO} T24 + mCh-GBP1 (C) or GBP1^{KO} HeLa + mCh-GBP1 (D) cells were treated with or without 1 μ g/mL aTc for overnight, and then infected with WT *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for actin. Percentages of *B. thailandensis* with actin tails were quantified. All bar graphs show 3 independent biological replicates, presented as mean $\pm$ SD. Two-way ANOVA with Tukey's multiple comparison tests were performed, with specific p-values indicated as follows: **, p < 0.01; ns, not significant.

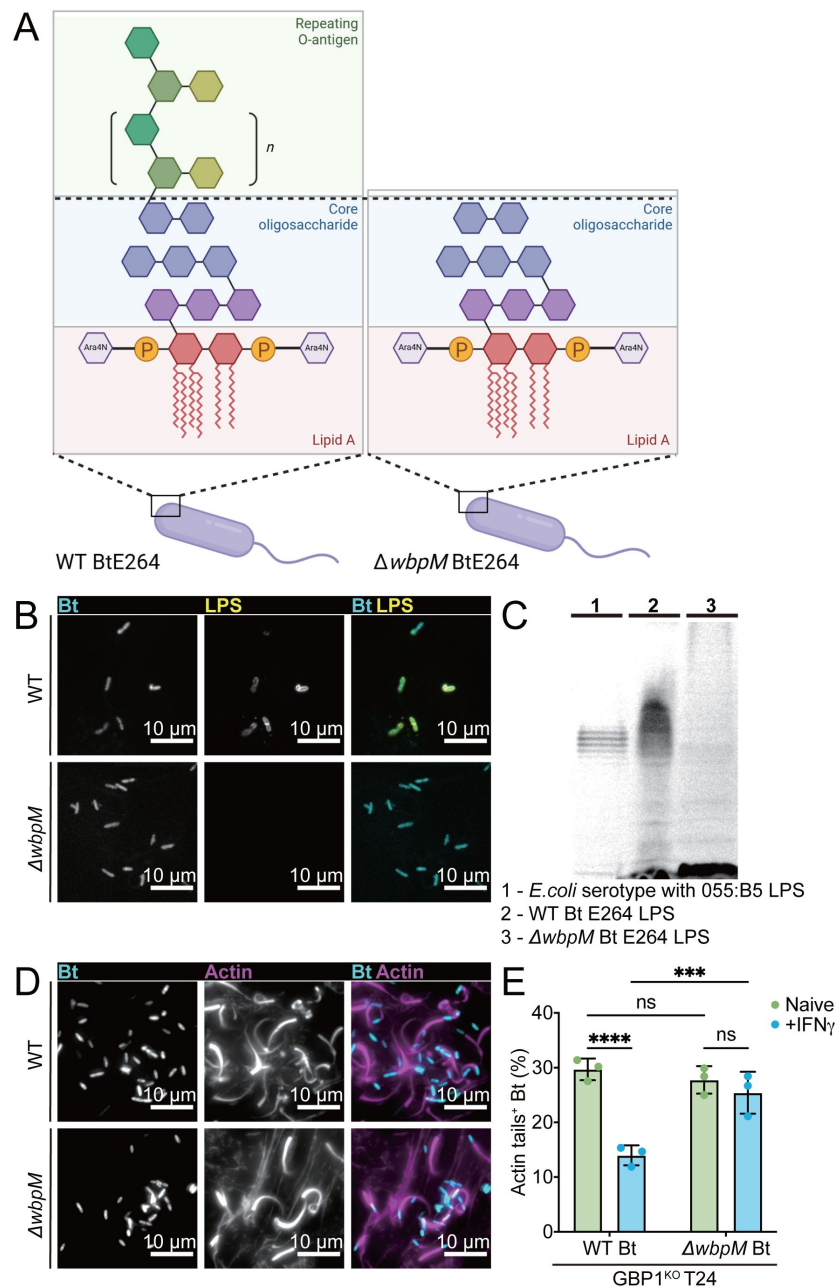


Fig. 3.

O-antigen is dispensable for *B. thailandensis* actin tail formation but promotes IFN γ -inducible host defense.

(A) Schematic depiction of WT *B. thailandensis* and $\Delta wbpM$ *B. thailandensis* LPS structure (B-E) Naïve GBP1^{KO} T24 cells were infected with WT or $\Delta wbpM$ *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for *B. thailandensis* LPS (B) and actin (D), and percentages of *B. thailandensis* with actin tails (E) were quantified. (C) SDS-PAGE analysis of LPS from *E. coli* serotype with 055:B5, WT *B. thailandensis* E264, or $\Delta wbpM$ *B. thailandensis* E264 with the Pro-Q Emerald 300 stain. All bar graphs show 3 independent biological replicates, presented as mean $\pm$ SD. Two-way ANOVA with Tukey's multiple comparison tests were performed, with specific p-values indicated as follows: ***, $p < 0.001$; ****, $p < 0.0001$; ns, not significant.

A genetic screen identifies GVIN1 as an inhibitor of *B. thailandensis* actin tail formation

We found that IFN γ -inducible pathways interfering with *B. thailandensis* actin tail formation are present in some (T24, U937, THP1) but absent from other (A549 and HeLa) human cell lines (Fig. 1 and S2). We therefore hypothesized that A549 and HeLa cells lack robust expression of specific ISGs that are required for the inhibition of actin tail formation. To identify candidate ISGs, we profiled gene expression in all five cell lines (T24, U937, THP1, A549, and HeLa) under naïve and IFN γ -primed conditions using bulk RNAseq and then used these data to curate a list of candidate genes that we could test for function (Fig. 4A). Based on our RNAseq data we identified 28 protein-coding ISGs that were induced at least 2 log-fold upon IFN γ priming in T24 cells and were expressed at least 2-fold higher in IFN γ -primed T24, U937, and THP1 cells relative to IFN γ -primed A549 or IFN γ -primed HeLa cells (Fig. 4A and S3A-B). To determine whether any of these candidate ISGs were involved in the GBP1-independent pathway, we conducted an arrayed functional genomics ‘mini screen’ in T24 GBP1^{KO} cells. For each candidate gene, individual wells seeded with T24 GBP1^{KO} cells were transfected with four independent siRNA oligonucleotides. We then primed the transfected cells with IFN γ overnight, infected the cells with *B. thailandensis*, and quantified the percentage of bacteria forming actin tails at 8 hours post infection (hpi). The number of cytosolic *B. thailandensis* forming actin tails inside IFN γ -primed T24 GBP1^{KO} cells significantly increased by approximately 2-fold relative to cells treated with siRNA control oligonucleotides for two targeted ISGs, namely *GBP6* (Gene ID: 163351) and *GVINP1* (GTPase, Very large interferon inducible pseudogene 1; Gene ID: 387751) (Fig. 4B and S3C). From hereon we will refer to *GVINP1* as *GVIN1*, as we will provide evidence for it to be a functional gene and not a pseudogene.

To independently validate whether GBP6 and GVIN1 play any role in limiting actin tail formation by *B. thailandensis*, we generated *GBP6* and *GVIN1* deletions in T24 cells using CRISPR technology (Fig. S4A-B). We found that IFN γ -primed T24 GBP6^{KO} cells restricted *B. thailandensis* actin tail formations at rates that were comparable to IFN γ -primed wild type T24 cells (Fig. S4C). Furthermore, GBP6 protein expression in naïve or IFN γ -primed T24 cells was below the level of detection by Western blotting (Fig. S4D). Collectively, these data suggested that GBP6 was a false-positive hit from our screen.

In contrast to GBP6^{KO} cells, GVIN1^{KO} cells allowed for increased rates of actin tail formation by *B. thailandensis* and showed incomplete restriction of actin tail formation upon IFN γ priming, thus mimicking the phenotype of GBP1^{KO} cells (Fig. 4C). Simultaneous deletion of both GVIN1 and GBP1 completely abolished the ability of IFN γ -primed T24 cells to limit actin tail formation by *B. thailandensis* (Fig. 4C). Together, these data indicate that GBP1 and GVIN1 are critical players in the execution of two independent host defense pathways targeting *B. thailandensis*.

GVIN1 coatomers require IFN γ -inducible co-factors to block actin tail formation

The *GVIN1* gene is intronless and expresses a transcript that is approximately 8,700 nucleotides in length. The *GVIN1* mRNA contains three potential open reading frames (ORFs). The longest uninterrupted ORF is 5,424 nucleotides in length and predicted to encode a protein of 1,808 amino acids with an expected molecular weight of 209 kDa (Fig. 5A), hereinafter simply referred to as GVIN1 protein. To gain insights into the potential function of GVIN1, we employed the protein structure prediction software AlphaFold3 (25, 26) and the structural alignment tool Foldseek (27). The predicted structure features an N-terminal domain that is connected to a Large GTPase (LG) domain via a flexible, mostly unstructured linker, followed by helical linker domain, a second LG domain, and a C-terminal domain (Fig. 5A-B and S5A-B). Because related GTPases of the

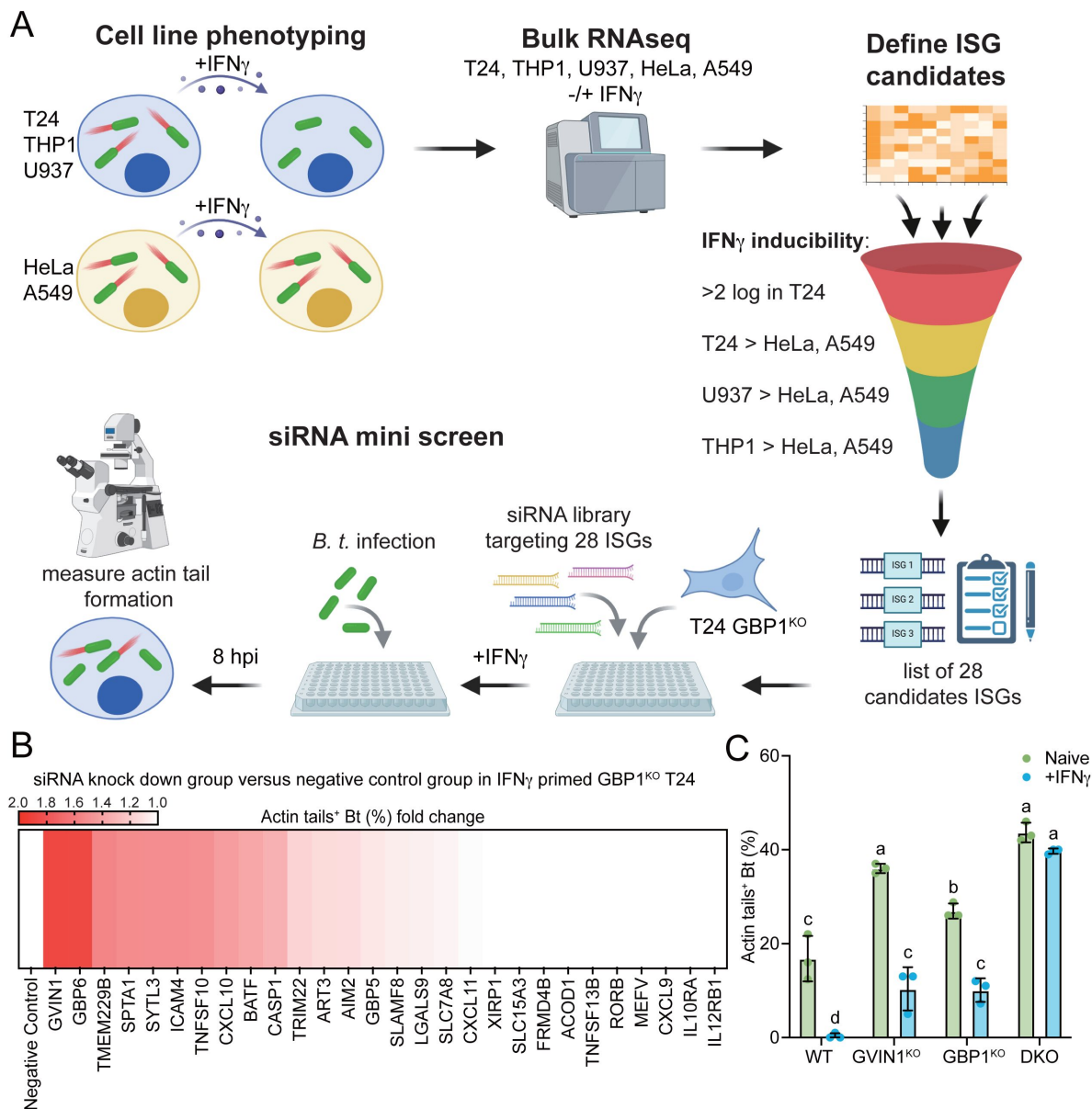


Fig. 4.

A genetic screen identifies GVIN1 as an inhibitor of *B. thailandensis* actin tail formation.

(A) Schematic depiction of the strategy for the siRNA mini screen. (B) 5 pmol of specific gene's siRNA transfected and 200 U/mL IFN γ primed GBP1^{KO} T24 cells were infected with WT *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for actin. Percentages of *B. thailandensis* with actin tails were quantified and showed as fold changes of %Actin-tail* *B. thailandensis* in the specific gene's siRNA knock down versus in the negative control siRNA transfected GBP1^{KO} T24 were shown in the heat map. (C) Naive or 200 U/mL IFN γ primed WT, GBP1^{KO}, GVIN1^{KO}, or GBP1^{KO} GVIN1^{KO} DKO T24 cells were infected with WT *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for actin. Percentages of *B. thailandensis* with actin tails were quantified. All bar graphs show 3 independent biological replicates, presented as mean $\pm$ SD. Two-way ANOVA with Tukey's multiple comparison tests were performed, with specific p-values indicated as follows: Different letters above columns represent there is significant difference, same letter above columns represents not significant.

GBP family form dimers upon GTP-binding and -hydrolysis (18 [↗](#)), we also modeled GVIN1 dimerization in AlphaFold3. Our predictions suggest that GVIN1 forms dimers via its LG domains (Fig. S5C-E [↗](#)), which is in line with dimerization of Mx proteins, IRGs, and GBPs (28 [↗](#)).

Dimeric GBP1 proteins insert into Gram-negative bacterial outer membranes to form coatomers and execute host defense (13 [↗](#), 29 [↗](#)–31 [↗](#)). To assess whether GVIN1 could operate in a similar fashion, we expressed a C-terminally HA-tagged GVIN1 protein under control of an aTc-inducible promoter in cell culture. Importantly, we found that expression of GVIN1-HA in T24 GVIN1^{KO} restored IFN γ -dependent restriction of *B. thailandensis* actin tail formation to levels comparable to WT cells (Fig. 5C [↗](#)). Ectopic expression of GVIN1-HA in T24 GVIN1/GBP1^{DKO} partially restored IFN γ -dependent restriction of actin tail formation comparable to the phenotype of GBP1^{KO} cells (Fig. 5D [↗](#)). These complementation experiments confirmed that loss of functional GVIN1 is responsible for elevated levels of *B. thailandensis* actin tail formation in GVIN1^{KO} cells. Next, we monitored the subcellular localization of ectopically expressed GVIN1-HA in *B. thailandensis* infected HeLa cells and observed that at 8 hpi approximately 50% of intracellular bacteria were encased by GVIN1-HA (Fig. 5E-F [↗](#)). This encapsulation morphologically resembled coatomers formed by GBP1 on *B. thailandensis* (Fig. 1A-B [↗](#) and (21 [↗](#))). Remarkably, GVIN1 coatomers formed only infrequently on the surface of the O-antigen deficient $\Delta wbpM$ *B. thailandensis* mutant (Fig. 5F [↗](#)), thus explaining why the $\Delta wbpM$ mutant strain is resistant to GBP1-independent, i.e. GVIN1-dependent, host defense (Fig. 3E [↗](#)).

Ectopically expressed GVIN1-HA formed coatomers around *B. thailandensis* both in unprimed and IFN γ -primed cells (Fig. 5F [↗](#)). However, a reduction in *B. thailandensis* actin tail formation required IFN γ priming in addition to ectopic GVIN1-HA expression both in HeLa and T24 cells (Fig. 5C-D [↗](#) and 5G-H), indicating that GVIN1 requires an IFN γ -inducible co-factor for host defense. Because ectopically expressed GVIN1-HA restricted bacterial actin tail formation in IFN γ -primed GBP1^{KO} HeLa as well as GVIN1/GBP1^{DKO} T24 cells (Fig. 5D [↗](#) and 5H [↗](#)), GBP1 can be excluded as an essential co-factor for GVIN1 function.

The *Burkholderia* actin nucleator BimA is displaced from the bacterial pole by both GBP1- and GVIN1-dependent host defense pathways

We previously demonstrated that GBP1 coatomers on the surface of *S. flexneri* in HeLa cells are sufficient to disrupt the polar localization of the *Shigella* actin nucleator IcsA (13 [↗](#)). In contrast to IcsA, we found that the polar localization of the *Burkholderia* actin nucleator BimA, tagged with the DYKDDDDK peptide (FLAG), remained unchanged in GBP1 coated bacteria residing within HeLa cells (Fig. S6A-B [↗](#)). These findings were expected considering that GBP1 coatomers are similarly insufficient to block *B. thailandensis* actin tail formation in HeLa cells (Fig. 1C [↗](#)). Next, we monitored BimA localization in T24 cells, which in contrast to HeLa cells dramatically reduce actin tail formation by *B. thailandensis* upon IFN γ priming (Fig. 1F [↗](#)). We observed that the number of bacteria with detectable BimA was dramatically diminished in IFN γ -primed WT T24 cells (Fig. 6A-B [↗](#)). In contrast to the outer membrane protein BimA, the BimA-interacting inner membrane protein BimC (32 [↗](#)) retained its polar localization in IFN γ -primed WT T24 cells (Fig. 6C-D [↗](#)). Polar BimA localization was partially restored in IFN γ -primed GBP1^{KO} and GVIN1^{KO} T24 cells (Fig. 6B [↗](#)). In GVIN1/GBP1^{DKO} the percentage of BimA-positive bacteria was identical between unprimed and IFN γ -primed cells (Fig. 6B [↗](#)). Together, these observations demonstrated that both the GBP1 and GVIN1 host defense pathways eliminate BimA from the bacterial outer membrane to block actin tail formation (Fig. 6E [↗](#)).

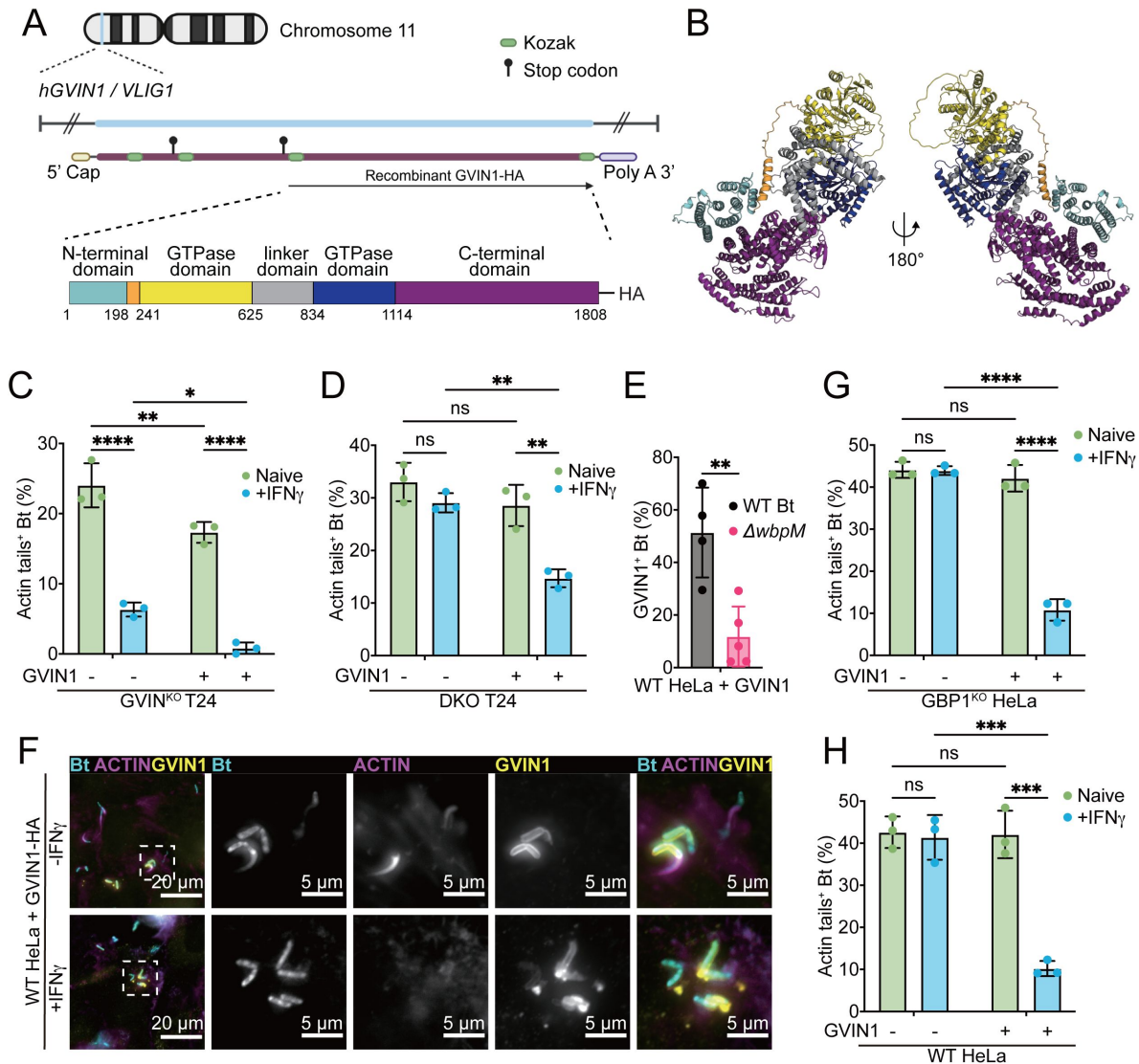


Fig. 5.

GVIN1 restricts *B. thailandensis* actin tail formation in an IFN γ dependent but GBP1 independent manner.

(A) Schematic depiction of the GVIN1 DNA, GVIN1 mRNA and the recombinant GVIN1 protein (C-terminal-HA-tagged protein translated from GVIN1 longest ORF). The domains were assigned by Foldseek of the predicted AlphaFold structure, the numbers reflect the position of the amino acid in the predicted protein sequence, starting with the start amino acid methionine of the longest ORF of GVIN1. (B) 3D structure model of the GVIN1 longest ORF, predicted by AlphaFold3. (C-H) Parental WT HeLa, GBP1^{KO} HeLa, GVIN1^{KO} T24, or GBP1^{KO} GVIN1^{KO} DKO T24 cells were transduced with aTc-inducible GVIN1-HA lentiviral construct. IFN γ primed and/or 1 μ g/mL aTc overnight-treated WT HeLa + GVIN1-HA (E), GBP1^{KO} HeLa + GVIN1-HA, GVIN1^{KO} T24 + GVIN1-HA, or GBP1^{KO} GVIN1^{KO} DKO T24 + GVIN1-HA were infected with WT *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for actin. Percentage of actin-tail-positive *B. thailandensis* were quantified (C, D, G, H). (F) Naïve or 200 U/mL IFN γ primed and/or 1 μ g/mL aTc overnight-treated WT HeLa + GVIN1-HA cells were infected with WT or $\Delta wbpM$ *B. thailandensis* at an MOI of 100. Cells were fixed at 8 hours post infection and stained for GVIN1-HA. Percentages of GVIN1-HA positive *B. thailandensis* were quantified. All bar graphs show 3 independent biological replicates, presented as mean $\pm$ SD. Two-way ANOVA with Tukey's multiple comparison tests were performed for (C), (D), (G), and (H); and unpaired t-test was performed for (F). Specific p-values indicated as follows: *, p < 0.05; **, p < 0.01; ***, p < 0.001; ****, p < 0.0001; ns, not significant.

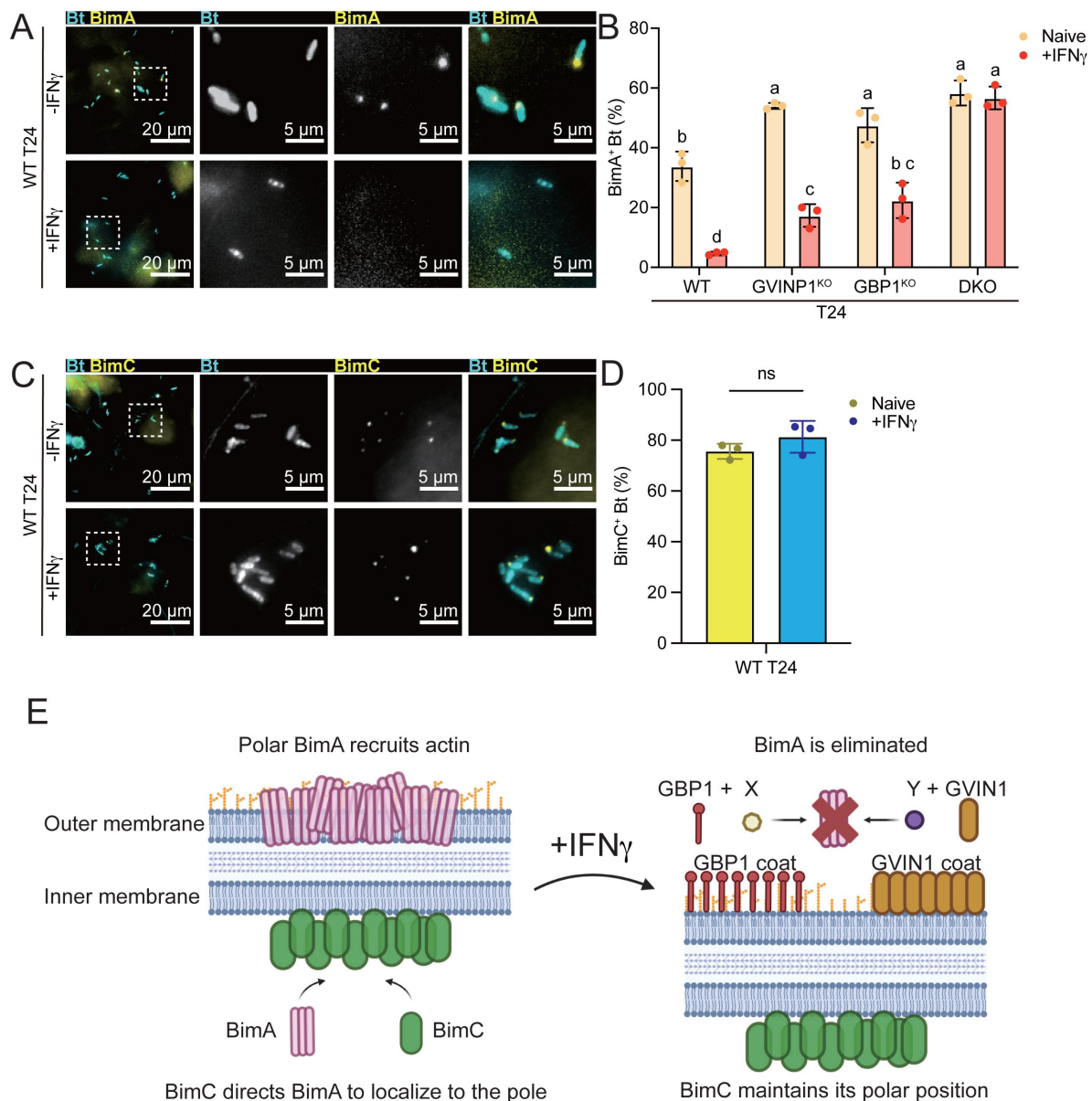


Fig. 6.

BimA is displaced from the bacterial pole via both GBP1- and GVIN1-dependent pathways, whereas BimC remains unaffected.

(A-B) Naïve or 200 U/mL IFN γ primed WT, GBP1^{KO}, GVIN^{KO}, or GBP1^{KO} GVIN^{KO} DKO T24 cells were infected with *B. thailandensis* expressing FLAG-tagged BimA at an MOI of 100. Cells were fixed at 8 hours post infection and stained for FLAG-BimA. Percentages of BimA positive *B. thailandensis* were quantified (B). (C-D) Naïve or 200 U/mL IFN γ primed WT T24 cells were infected with *B. thailandensis* expressing GFP-tagged BimC at an MOI of 100. Cells were fixed at 8 hours post infection. Percentages of GFP-BimC positive *B. thailandensis* were quantified (D). All bar graphs show 3 independent biological replicates, presented as mean $\pm$ SD. Two-way ANOVA with Tukey's multiple comparison tests were performed for (B), and unpaired t-test was performed for (D). Specific p-values indicated as follows: (B) columns not sharing any letters means significantly different ($p < 0.05$); (D) ns, not significant. (E) Schematic depiction of the model, Polar BimA that recruits actin is eliminated by the GBP1 pathway and GVIN1 pathway together in IFN γ primed T24 cells.

Discussion

IFN-inducible GTPases are potent executioners of cell-autonomous immune programs targeting viral, bacterial, fungal, and protozoan intracellular pathogens. Previous work focused on the function of three of these GTPases families, the Mx proteins, the IRGs, and the GBPs (17–20, 33, 34). While *GVIN* genes encoding members of the fourth family of IFN-inducible GTPases are present in most vertebrate genomes and have previously been shown to be expressed in mice (35), their biological function has remained enigmatic. Our current work demonstrates that human *GVIN1*, the sole human member of the *GVIN* family, forms a coatomer around *B. thailandensis* and interferes with bacterial actin tail formation. Our observations thus establish *GVIN* GTPases as a new class of host defense proteins.

Previous work demonstrated that human *GBP1* can form coatomers on the surface of numerous Gram-negative bacteria to combat intracellular infections (13, 29–31). The *GBP1* coatomer acts like an emulsifier and disrupts the chemo-mechanical barrier established by the polysaccharide O-antigen portion of bacterial LPS (13). In concert with additional antimicrobial molecules the emulsification of the LPS barrier by *GBP1* can promote the killing of encased bacteria (13, 36). On its own the *GBP1* coatomer appears to be neither bacteriostatic nor bactericidal (21), yet its detergent-like activity is sufficient to increase the diffusion rate of the *Shigella* outer membrane protein *IcsA*, thereby interfering with *IcsA*'s critical role in nucleating actin at one bacterial pole and establishing actin-based motility (13). In contrast to the *Shigella* *IcsA* protein, we show here that the functionally related *BimA* protein of *B. thailandensis* is impervious to the disruption of the O-antigen layer. Instead, one or more unidentified *GBP1* co-factors are required for *GBP1* coatomers to block actin tail formation by *B. thailandensis*.

Like *GBP1* coatomers, *GVIN1* coatomers require additional co-factors to effectively block *B. thailandensis* actin formation. The nature of these co-factors will need to be determined in future studies. However, because ectopic expression of *GVIN1* but not *GBP1* in HeLa cells reduces *B. thailandensis* actin tail formation, we can already conclude that HeLa cells express essential *GVIN1* but not *GBP1* co-factors. Accordingly, we can infer that *GBP1* and *GVIN1* use distinct sets of co-factors. Despite these clear differences, both *GBP1* and *GVIN1* coatomers, in cooperation with their respective co-factors, function similarly by removing *BimA* from the bacterial outer membrane to disrupt actin tail formation by *B. thailandensis*. The polar localization of the inner membrane protein *BimC* on the other hand is undisturbed, suggesting that both *GBP1* and *GVIN1* coatomers specifically disrupt modalities of the outer bacterial membrane.

The basic principles underlying the assembly of *GBP1* coatomers have been established. Upon GTP binding and hydrolysis *GBP1* monomers undergo conformational changes and form dimers (37, 38). These dimers can polymerize (39), a process that is aided by the presence of bacterial LPS (13, 40). The polymers dock to the surface of Gram-negative bacteria, penetrate the bacterial polysaccharide barrier, and release thousands of *GBP1* dimers into the bacterial outer membrane to form a *GBP1* coat on the bacterial surface (13, 18, 30). Although our structural modelling suggests that *GVIN1* GTPases may also form dimers, whether *GVIN1* coatomers assemble through a multi-step process that is analogous to *GBP1* coatomer formation will require future investigations.

Our studies did not systematically explore whether *GVIN1* targets intracellular pathogens other than *B. thailandensis*, although this seems likely. Remarkably, we found that O-antigen deficient *B. thailandensis* rough mutants had become resistant to *GVIN1*-mediated host defense, which suggests that *GVIN1* either detects moieties present in *B. thailandensis* O-antigen or that O-antigen helps to stabilize the *GVIN1* coat. This observation is also noteworthy in the context of reports showing that clinically important *Burkholderia* species such as *Burkholderia cenocepacia* and

Burkholderia multivorans isolated from chronically infected cystic fibrosis patients tended to acquire mutations that alter or remove O-antigen (41 [↗](#)–43 [↗](#)). While speculative, changes in O-antigen composition and expression may provide these clinical *Burkholderia* isolates with a formidable fitness advantage by obtaining resistance to GVIN1.

ISGs play crucial roles in executing cell-autonomous immune responses. However, only a limited number of these ISGs have been thoroughly characterized, and there is little understanding of how they functionally interact with one another. Here, we characterize GVIN proteins as novel executioners of cell-autonomous immunity. Like related GBP1 proteins, human GVIN1 proteins encase their bacterial target, spotlighting coatomer formation as a fundamental mechanism of antibacterial cell-autonomous defense. Furthermore, we show that both GBP1 and GVIN1 coatomers collaborate with other ISGs, highlighting ISG synergy as a principal operating mechanism of antimicrobial coatomer activity. Elucidating the formation and defensive role of GVIN1 coatomers, along with potential microbial evasion strategies, could offer valuable insights into host-pathogen interactions and aid in developing novel therapies for infectious diseases.

Materials and methods

Cell lines and culture

HeLa (ATCC #CCL-2), A549 (ATCC #CCL-185), and 293T (ATCC #CRL-3216) cell lines were cultivated in Dulbecco's Modified Eagle Medium (DMEM, Gibco) supplemented with 10% heat inactivated fetal bovine serum (Omega Scientific), 1% MEM Non-Essential Amino Acids Solution (MEM NEAA, Gibco), and 55 μ M 2-Mercaptoethanol (Gibco). T24 (ATCC #HTB-4) cells were cultivated in McCoy's 5A (Modified) Medium (Gibco) supplemented with 10% heat inactivated fetal bovine serum. THP-1 (ATCC #TIB-202) and U-937 (ATCC #CRL-1593.2) cells were cultivated in Roswell Park Memorial Institute 1640 (RPMI 1640, Gibco) supplemented with 10% heat inactivated fetal bovine serum, and 55 μ M 2-Mercaptoethanol (Gibco). THP-1 and U-937 monocytes were differentiated to macrophages by incubation with 50ng/mL phorbol 12-myristate 13-acetate (PMA, Sigma-Aldrich) for 48 hours. All cell lines were grown at 37 °C with 5% CO₂. All cell lines were authenticated by the Duke University DNA Analysis Facility using GenePrint10 (Promega) and were routinely tested for mycoplasma.

Knock out cell lines

The knockout (KO) GBP1, GVIN1, and GBP6 T24 cell lines were generated by the Duke Functional Genomics core using CRISPR/Cas9 technology. CRISPR sgRNAs targeting GBP1, GVIN1, and GBP6 were designed using ChopChop and ordered as modified synthetic sgRNAs from Synthego (44 [↗](#)). For GBP1, the sgRNA sequence 5'-TGCCATTACAGCCTATGG-3' was used. For GVIN1, the sgRNA sequence 5'-TGTACATTGTGGCTGTAGCG-3' was used. For GBP6 the sgRNA sequences 5'-TCTGGCAGTGAATTATGTAG-3' was used. 5.6×10^4 T24 cells were electroporated with 6 pmol TrueCut Cas9 protein v2 (ThermoFisher Scientific) complexed with 36 pmol sgRNA using the Neon system (ThermoFisher Scientific) with the following settings: 1600 V, 10 ms, 3 pulses. Cells were recovered and expanded, followed by PCR sequencing and analysis using Inference of CRISPR Edits (ICE) to determine KO efficiencies for each individual sgRNA (45 [↗](#)). For each KO cell line, single clones were isolated by plating at limited dilution in 96 well plates. Clones were expanded for approximately 10-14 days before screening by PCR-sequencing to confirm that the edit resulted in a frame-shift mutation. The GBP1^{KO} HeLa cell line was described previously (13 [↗](#), 46 [↗](#)). The CASP4^{KO} THP-1 cell line (47 [↗](#)) was a gift from Dr. Veit Hornung, University of Munich, Germany, and Dr. Eva Frickel, University of Birmingham, United Kingdom.

Bacterial strains

Wild-type *B. thailandensis* strain E264 carrying a GFP expression construct was a gift from Dr. Edward Miao, Duke University, USA. For the BimA studies, a *B. thailandensis* strain E264 was used with BimA chromosomally replaced with Flag-BimA (10 [DOI](#)), which was a gift from Dr. Matthew D. Welch, University of California, Berkeley, USA. For the BimC studies, wild-type *B. thailandensis* strain E264 was transformed with pME6032-GFP-BimC (32 [DOI](#)), a gift from Dr. Feng Shao, National Institute of Biological Sciences, Beijing, China. The *wbpM* (BTH_I1483) gene was deleted from the native locus of *B. thailandensis* by allelic exchange using homologous recombination. Regions (~500 bp) flanking *wbpM* with a chloramphenicol resistance cassette (Cm^r) in the middle were cloned into a pEXKm5 suicide vector (48 [DOI](#)), which was kindly provided by Dr. Peggy Cotter, University of North Carolina, Chapel Hill, USA. The recombinant plasmid was transformed into *B. thailandensis*. Briefly, an overnight culture of *B. thailandensis* was back-diluted and grown in minimal media, M63 for 4 hours (M63 5X salt solution was made by dissolving 15 g KH_2PO_4 , 35 g KHPO_4 , 10 g $(\text{NH}_4)_2\text{SO}_4$, and 2.5 mL of a 16 nM FeSO_4 in 1 L of water and the pH was adjusted to 7.0; to make 1 L of the final M63 medium, 10 mL of 1% casamino acids was added to 780 mL water and autoclaved. Sterile 200 mL of the M63 5X salt solution, 10 mL of 20% glucose, 1 mL of 1 M MgSO_4 , and 8 mL of 50% glycerol was added). *B. thailandensis* mixed with linearized pEXKm5 suicide vector containing *wbpM* flanking sequences were grown overnight in minimal media. The next day, the culture was pelleted and spread on agar plates to select for chloramphenicol resistance. Double crossover was confirmed by PCR using Cm^r -F (5'-TCAGTCAGCGCCGCCACGAGGCCCTTTTCGTCTTCGAATAA-3') and Cm^r -R (5'-AGTCACTGCGGCCGCTTGCGCTCACTGCCCCGCTTTCCAG-3') as well as *wbpM*-Flnk-F (5'-TCGATGACCGTCGCGGATTGC-3') and *wbpM*-Flnk-R (5'-ATGGTCGCGTTTCATGATGTTTCGC3') primer sets. *S. flexneri* $\Delta ipaH9.8$ strain carrying pGFPmut2 was previously reported (21 [DOI](#)). Bacterial cultures were supplemented with the following antibiotics when required: carbenicillin (50 $\mu\text{g/mL}$), kanamycin (50 $\mu\text{g/mL}$), chloramphenicol (25 $\mu\text{g/mL}$), gentamicin (25 $\mu\text{g/mL}$). Antibiotic concentrations were maintained throughout infection experiments.

LPS visualization

Overnight grown bacterial cultures were lysed with RIPA buffer (Thermo Fisher Scientific, 89900) and checked for protein concentration using bicinchoninic acid (BCA) assay kit (Thermo Fisher Scientific). Aliquots of 20 μg samples were diluted in Laemmli buffer and applied to a 4–20% Mini-PROTEAN® TGX™ Precast Protein Gels (Biorad, 4561094). The gel was processed with the Pro-Q Emerald 300 LPS stain kit (Thermo Fisher Scientific, P20495), according to the manufacturer's protocol. The LPS bands were visualized under UV light using the Azure 500 imaging system (Azure Biosystems).

Bacterial infection

Cell lines (T24, HeLa, A549, THP-1, and U-937) were seeded in 24-well plate (immunofluorescence and CFU assays) or 96-well plates (LDH assay) and treated with 200 U/ml IFN γ (Sigma-Aldrich) overnight before infection. For *B. thailandensis* infections, all *B. thailandensis* strains were grown at 37°C on lysogeny broth (LB) agar plates. 5 mL of LB supplemented with appropriate antibiotics was inoculated with one *B. thailandensis* colony and incubated overnight at 37°C with shaking. On the day of infection, the overnight culture was diluted 1:50 in 5 mL of fresh LB with antibiotics and grown at 37°C with shaking for 3–4 hours until the optical density at 600 nm (OD₆₀₀) reached 0.6–0.8. The bacterial cultures were pelleted by centrifugation at 5000 rpm for 3 minutes at room temperature (RT), resuspended in 1 mL of phosphate-buffered saline (PBS, Gibco), and centrifuged again under the same conditions. The washed bacteria were then resuspended in cell culture media containing antibiotics and further diluted in cell culture media to the desired concentration. T24, HeLa and A549 cells were infected at a multiplicity of infection (MOI) of 100, THP-1 and U-937 cells were infected at a MOI of 20. Following centrifugation at 700 $\times$ g for 10 minutes at RT, infected

cells were incubated at 37°C with 5% CO₂. At 4 hpi, cells were washed twice with Hank's Balanced Salt Solution (HBSS, Gibco) to remove extracellular bacteria. Cells were then incubated in media containing chloramphenicol for 2 hours to eliminate remaining extracellular bacteria. Afterwards, the culture media was replaced with antibiotic-free media. The infection was terminated at 8 hpi for T24, HeLa, and A549 cells, and at 6 hpi for THP-1 and U-937 cells. For *S. flexneri* infections, bacteria were grown at 37°C on tryptic soy broth (TSB) agar plates containing 0.01% Congo red. 5 mL of TSB supplemented with appropriate antibiotics was inoculated with one *S. flexneri* colony and incubated overnight at 37°C with shaking. On the day of infection, the overnight culture was diluted 1:50 in 5 mL of fresh TSB with antibiotics and grown at 37°C with shaking for about 2 hours until the OD₆₀₀ reached 0.5-0.7. The bacterial cultures were pelleted by centrifugation at 5000 rpm for 3 minutes RT, resuspended in the 990 µL PBS supplement with 10 µL of 0.1 mg/mL polyD-lysine (Sigma-Aldrich) solution, and then incubated at 37°C with shaking for 15 minutes. The *S. flexneri* were pelleted again by centrifugation at 5000 rpm for 3 minutes RT. The washed bacteria were then resuspended in cell culture media containing antibiotics and further diluted in cell culture media to the desired concentration. T24, HeLa cells were infected at an MOI of 10, followed by centrifugation at 700 × g for 10 minutes at RT. Infected cells were incubated at 37°C with 5% CO₂. 30 minutes post-infection, cells were washed twice with HBSS to remove extracellular bacteria. Cells were then incubated in culture media containing gentamicin to eliminate extracellular bacteria. The infection was terminated at 2 hours and 30 minutes post infection for T24 and HeLa cells.

Expression constructs and lentivirus production

An anhydrotetracycline-inducible GVIN1 plasmid was constructed using Gateway assembly (Thermo Fisher) into the pMVP cloning system. The pMVP Cloning System was a gift from Christopher Newgard (Addgene kit #1000000155) (49). The third ORF of GVIN1 sequence without stop codon flanked by attB4 and attB3r recombination sites was amplified via PCR using primers, CloneGVINP1F3 (5'-GGGGACAACCTTTCTATACAAAGTTGCCACCATGCAGTGGATCTTCCAGTC-3'), and CloneGVINP1nostopR1 (5'-GGGGACAACCTTTATTATACAAAGTTGTCTGTTTTTCAAGTCTTCAAGCAC-3') with genomic DNA from T24 cells as the template. The PCR product was cloned into the pDONR221-P4r-P3r vector (NovoPro) using BP Clonase™ II (Thermo Fisher), generating a GVIN1 entry clone. Subsequently, the following entry clones were assembled in an LR recombination reaction: pDONR221-R4-R3-GVIN1, Tre promoter entry clone (Addgene Plasmid #121691), HA tag-polyA + CMV-TETa-polyA entry clone (Addgene Plasmid #121802). Entry clone plasmids were added with the pMVP/Lenti/Neo-DEST vector (Addgene Plasmid #121850), and LR Clonase™ II Plus enzyme mix (Thermo Fisher) together to generate the final tetracycline-inducible HA-tagged GVIN1 expression lentiviral construct (pMVP-lenti-neo-largeORFGVIN1-HA). Lentiviral particles were produced by transfecting lentivirus plasmids with packaging vectors into HEK293T cells using the TransIT™ 293 transfection reagent (Mirus Bio) according to the manufacturer's instructions. Each well of a 6-well plate containing 293T cells was transfected with 1 µg of plasmid, 750 ng of pSPAX2, and 250 ng of VSVG. At 24 hours post-transfection, the media were replaced with DMEM. The supernatants containing lentiviral particles were collected at 48- and 72-hours post-transfection. Supernatants were pooled and filtered through 0.45-µm nylon filters (Corning). For lentiviral transduction, 293T cells were diluted to a concentration of 3.33 × 10⁴ cells/mL in media containing 10 µg/mL polybrene. A total of 500 µL of lentivirus was added to each well of a 6-well plate containing recipient cells, followed by the addition of 1.5 mL of diluted cells. The plates were incubated at 37°C with 5% CO₂ for 48 hours. Post transduction, cells were selected in media containing geneticin (Gibco, 500 µg /mL for HeLa, 1000 µg /mL for T24). An anhydrotetracycline-inducible GBP1 plasmid was described previously (13, 40, 46). GBP1^{KO} HeLa cells and GBP1^{KO} T24 cells were stably transduced with this anhydrotetracycline-inducible mCherry-hGBP1 expression system, as described above.

Immunofluorescence microscopy

Post-infection, cells were fixed in 4% paraformaldehyde in PBS at RT for 20 minutes. Fixed cells were permeabilized with 0.25% Triton X-100 in PBS for 10 minutes at RT. Permeabilized cells were then blocked with blocking buffer (5% bovine serum albumin and 0.3 M glycine in PBS) at RT for 45 minutes. Samples were incubated with primary antibodies diluted in blocking buffer at RT for 1 hour. Primary antibodies were diluted in blocking buffer as follows: rabbit recombinant monoclonal anti-GBP1 (1:150; Abcam, ab131255); mouse monoclonal ANTI-FLAG® M2 (1:1000; Sigma-Aldrich, F1804); mouse monoclonal anti-HA tag [HA.C5] (1:500; Abcam, ab18181); recombinant rabbit monoclonal Anti-LPS [CK3 (BP1 7F7)] (absolute antibody, Ab03071-23.0). Following primary antibody incubation, samples were washed three times in 0.05% Triton X-100 in PBS for 5 minutes at RT. Subsequently, samples were incubated with secondary antibodies diluted in blocking buffer at RT for 1 hour. Secondary antibodies and phalloidin were prepared in blocking buffer as follows: Alexa Fluor™ 568-conjugated donkey anti-rabbit IgG (H+L; 1:1000; Thermo Fisher, A10042), Alexa Fluor™ 405-conjugated goat anti-mouse IgG (H+L; 1:1000; Thermo Fisher, A31553), Alexa Fluor™ 568-conjugated goat anti-mouse IgG (H+L; 1:1000; Thermo Fisher, A-11004), Alexa Fluor™ 488-conjugated goat anti-rabbit IgG (H+L; 1:1000; Thermo Fisher, A-11008), Alexa Fluor™ 660 phalloidin (1:400; Thermo Fisher, A22285) for F-actin visualization. Cells were washed four times in 0.05% Triton X-100 in PBS for 5 minutes at RT. Coverslips with cells were mounted on glass slides using a mixture of Mowiol 4-88 and para-phenylenediamine antifading agent (9:1 ratio) to preserve fluorescence. Images were acquired using a Zeiss Axio Imager microscope equipped with a 63×/1.4 NA oil immersion lens. For each condition, at least six fields of view were captured to analyze at least 100 bacteria. Z-stack images were taken at 1 µm intervals. Bacteria with protein signals surrounding at least 50% of its visible surface were counted as targeted.

Western blotting

For cell lysis and protein extraction, cells were washed twice with Hank's Balanced Salt Solution (HBSS) and lysed in RIPA buffer (Sigma-Aldrich) supplemented with 4 U/ml DNase I (New England Biolabs) and protease inhibitor cocktail (Sigma-Aldrich). Lysates were incubated at 4°C for 30 minutes, followed by centrifugation at 20,000 × g for 10 min at 4°C. Protein concentrations were quantified using a bicinchoninic acid (BCA) assay kit (Thermo Fisher Scientific) and normalized prior to analysis. Normalized samples were mixed with Laemmli buffer containing 5% β-mercaptoethanol and denatured by heating at 95°C for 10 min. Protein samples were resolved on 4–20% Mini-PROTEAN TGX Stain-Free precast gels (Bio-Rad) using Tris-glycine running buffer. Proteins were transferred to PVDF membranes using the Tank Transfer System or Trans-Blot Turbo Transfer System (both Bio-Rad). Primary antibodies diluted in blocking solution (5% non-fat dry milk in 0.1% TBS-T) were incubated with membranes overnight at 4°C. After three washes with TBS-T (0.1% Tween-20 detergent in tris-buffered saline), membranes were incubated with species-matched horseradish peroxidase (HRP)-conjugated secondary antibodies for 1 hour at room temperature. After four washes with TBS-T, protein bands were visualized using Amersham ECL Prime Western Blotting Detection Reagent (Cytiva) and imaged on an Azure 500 imaging system (Azure Biosystems).

LDH assay

Cell death was evaluated using the lactate dehydrogenase (LDH) release assay (CytoTox-ONE™ Homogeneous Membrane Integrity Assay (Promega)) to measure cell membrane damage. Briefly, an equal volume (50–100 µL) of CytoTox-ONE™ reagent was added to each well in a 96 well plate with cell samples post-infection. The reaction mixture was incubated in the dark at RT for 10 min to allow enzymatic conversion of resazurin to resorufin by released lactate dehydrogenase (LDH). Fluorescence intensity was measured using a plate reader (Biotek Synergy H1, Agilent).

RNA sequencing

T24, HeLa, A549 cells were each plated in two wells of a 6 well plate (Corning) and allowed to grow overnight in standard media and culture conditions (described above). After 24 hours, media was replaced with fresh standard media (control unprimed well) or media containing 100 U/ml IFN γ (primed well). Six hours after priming, RNA was harvested for analysis. THP1 and U937 cells were differentiated into macrophages by treating with 50 ng/ul PMA in standard media for 48 hours and then cultured for another 48 hours in standard media without PMA. Then, media was replaced with fresh standard media (control unprimed well) or media containing 100 U/ml IFN γ (primed well). Six hours after priming, RNA was harvested for analysis. RNA was isolated using RNeasy Mini Kit (Qiagen) following manufacturer's instructions. The RNA library was prepared using KAPA mRNA Hyperprep and sequenced on the Illumina NovaSeq 6000. RNA-seq data was processed using the fastp toolkit (50 [↗](#)) to trim low-quality bases and sequencing adapters from the 3' end of reads, then mapped to GRCh38 (downloaded from Ensembl, version 106) (51 [↗](#)) using the STAR RNA-seq alignment tool (52 [↗](#)) and reads aligning to a single genomic location were summarized across genes. For genes having an overlap of at least 10 reads, gene counts were normalized, and differential expression was carried out using the DESeq2 (53 [↗](#)) and Bioconductor (54 [↗](#)) package implemented for the R programming environment. Consistent with the recommendation of the DESeq authors, independent filtering (55 [↗](#)) was utilized prior to calculating adjusted p-values (56 [↗](#)) and moderated log2 fold-changes were derived using the ashR package (57 [↗](#)).

siRNA knockdown

Gene knockdown in T24 cell lines was achieved by transfecting cells with siGENOME siRNA Reagents (Horizon) using Lipofectamine $\text{\textcircled{R}}$ RNAiMAX Reagent (Thermo Fisher Scientific) following the manufacturer's instructions. In brief, T24 cells were seeded in 24-well plates to achieve 60–80% confluency at the time of transfection. siRNA-lipid complexes were prepared in Opti-MEM $\text{\textcircled{R}}$ Reduced Serum Medium (Gibco). The siRNA and lipofectamine reagents were diluted separately in Opti-MEM $\text{\textcircled{R}}$ medium before mixing the dilutions in a 1:1 volume ratio. Following incubation at room temperature for 5 minutes to allow formation of the siRNA-lipid complexes, the reaction mix was added to the T24 cells. The final transfection mixture for each well contained 5 pmol siRNA and 1.5 μ L Lipofectamine in 50 μ L siRNA-lipid Opti-MEM $\text{\textcircled{R}}$ complex. After 48 h, transfected cells were used for the infection screening. The siRNA sequences used are listed in [table 1 of our supplementary data](#) [↗](#).

Protein structure prediction and modelling

Structural models of GVIN1 were predicted with the AlphaFold server (<https://alphafoldserver.com> [↗](#)) using the AlphaFold3 algorithm (26 [↗](#)). Therefore, the protein sequence was provided as input, and for the dimer model, guanosine triphosphate and divalent magnesium cations were chosen as ligands and ions, respectively. To define the domain structure, structural alignments were done with Foldseek (27 [↗](#)). Structural analysis and comparison were conducted using the PyMOL molecular graphics system (version 3.1.3.1, Schrödinger). PAE plots were visualized with UCSF ChimeraX (version 1.9).

Statistical analysis

All graphical representations and statistical analyses were performed using GraphPad Prism 10. All experimental data were derived from at least 3 independent experiments and are presented as mean $\pm$ standard deviation (SD) in bar/line graphs. Parametric statistical comparisons were conducted using Two-tailed unpaired Student's t-test for pairwise comparisons and Two-way ANOVA with Tukey's multiple comparisons test for factorial experimental designs. p-values are indicated as follows: *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

Data storage

The RNA-seq data discussed in this publication have been deposited in NCBI's Gene Expression Omnibus (58 [link](#)) and will be accessible through GEO Series accession number GSE288527 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE288527> [link](#)). The numerical values of all other quantified data depicted in data panels in this manuscript will be made openly available in the Digital Repositories at Duke at the time at which the final version of the manuscript is published. Digital object identifier will be provided.

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Additional files

Supplementary data. [link](#)

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Reviewer #1 (Public review):

This is a very elegant and convincing study. Using systematic screening of actin tail formation in two bacterial strains and employing a panel of CRISPR-CAS ko cell lines, the authors identify a novel dynamin-related GTPase GVIN, which forms an oligomeric coat around an intracellular Burkholderia strain. The bacterial O-antigen LPS layer is required for the formation of the GVIN coat, which disturbs the polar localization of the bacterial actin-polymerizing BimA protein.

I am not an expert in infection studies, but the experiments appear to be of high quality, the figures are well prepared, and clean and statistically significant results are provided. I have no criticism of the presented approaches.

The identification of a novel GBP1-independent pathway targeting intracellular bacteria is not only of fundamental importance for the immunity field but also of high interest to researchers in other areas, for example, evolutionary or structural biologists.

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Reviewer #2 (Public review):

Summary:

The authors wanted to investigate how cells defend against intracellular pathogens, such as *Shigella* and *Burkholderia* species, that co-opt the host actin machinery to spread from cell-to-cell. Previous work has identified IFN γ -inducible GTPase of the Guanylate Binding Protein (GBP) family in cytosolic defence against Gram-negative bacteria. By forming a coat around *Shigella*, human GBP1 suppresses actin-based motility by displacing IcsA, which is the actin-polymerising virulence factor present at bacterial poles. In addition, GBP1 recruits the cytosolic LPS-sensor, caspase-4, to the bacterial surface, which results in the removal of bacterial replicative niches via pyroptotic cell death. Here, they followed up their finding that GBP1 can reduce actin-based motility of *Shigella* in HeLa cells and, surprisingly, fails to do so during *Burkholderia* infection. In contrast, in T24 bladder epithelial cells, GBP1 is competent in blocking *Burkholderia* actin-tails. They therefore wanted to identify the GBP1-independent factor that blocks actin-based motility in IFN γ -treated cells that is absent in HeLa cells.

Major strengths and weaknesses of the methods and results:

The authors report a second IFN γ -dependent pathway involving the protein product of the gene *GVIN1*, which was previously thought to be a pseudogene. *GVIN1* (GTPase, very large interferon inducible 1) is thus the first human member of this family of ~250 kDa putative GTPases to be demonstrated to be functional and have potential antimicrobial roles. The discovery that *GVIN1* is indeed functional, forms coats on *Burkholderia* in an LPS O-antigen-dependent manner, and limits actin-dependent motility are the main strengths of this paper. The authors use CRISPR/Cas9-based knockouts in HeLa and T24 cells, and complement them to demonstrate that GBP1 and *GVIN1* are both required to inhibit actin-based motility.

An appraisal of whether the authors achieved their aims and whether the results support their conclusions:

The authors achieved their main goals through well-planned experiments and unbiased screens. They succeeded in finding the factor that blocks actin-based motility independently of GBP1. This is driven by *GVIN1*, which coats bacteria and limits actin-tail formation by reducing the expression of BimA through currently unknown mechanisms. Further, they found that an O-antigen mutant can escape coating by *GVIN1*, indicating the requirement for these polysaccharides in *GVIN1*-dependent bacterial sensing. However, the authors have not investigated whether *GVIN1*, which has two GTPase-domains, does indeed have GTPase activity and whether *GVIN1* and GBP1 together completely block cell-to-cell spread by *Burkholderia* and thereby restrict bacterial numbers over the infection time course. They also do not show whether GBP1 and *GVIN1* target the same bacterial cell or different populations of bacteria.

A discussion of the likely impact of the work on the field, and the utility of the methods and data to the community:

This work uncovers the antimicrobial actions of a member of yet another family of IFN γ -induced GTPases, which potentially acts against other intracellular pathogens. *GVIN1* appears to operate independently and in parallel to GBP1, pointing to the breadth and complexity of the IFN γ -inducible GTPase families.

<https://doi.org/10.7554/eLife.106896.1.sa1>

Reviewer #3 (Public review):

Summary:

Here, Guo et al. (2025) propose that the IFN-induced GTPase *GVIN1* forms a coat on cytosolic *Burkholderia thailandensis*, blocking actin tail formation through a mechanism analogous to

GBP1-mediated restriction of *Shigella* motility.

Their study was prompted by the intriguing observation that IFN γ priming and GBP1 coat formation fail to inhibit *B. thailandensis* actin-based motility in HeLa cells, yet IFN γ restricts the motility of *Burkholderia* in T24 cells. Further investigation revealed that IFN γ restricts *B. thailandensis* motility in T24 cells via both GBP1-dependent and -independent mechanisms, suggesting that HeLa cells lack a critical GBP1 co-factor required to inhibit actin tail formation.

To identify the GBP1-independent mechanism, the authors performed an siRNA screen of interferon-stimulated genes (ISGs) and identified GVIN1, a large IFN-induced GTPase, as essential for restricting *B. thailandensis* motility. To identify the GBP1-independent mechanism, perform a knock-down screen for ISGs and find that the loss of GVIN, a very large IFN-induced GTPase, results in higher actin tail-positive *B. thailandensis* in T24 cells. They further demonstrate that GVIN forms coats on the surface of *B. thailandensis*, which prevent the polar localization of BimA and thus actin tail formation. In summary, the data reveal two independent IFN γ -induced pathways that restrict bacterial motility: one GBP1-dependent and the other GVIN1-dependent, each relying on distinct host co-factors.

Global assessment:

This is a well-executed study that convincingly demonstrates how GVIN1 restricts the actin-based motility of *B. thailandensis* through the assembly of coatomers. The results are clearly described, the manuscript is easy to follow, and the data are overall compelling and well presented. I have only a few suggestions on how the manuscript could be further improved.

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